



ELECTRIC TRAINS

Electric trains run on electricity that is provided by an overhead line or an electrified rail. They do not give off smoke and can travel very fast.



FASTEST TRAINS

An early record-setting train was *Stephenson's Rocket*, at 46 km/h (29 mph). In the 1960s, Japan opened the world's first high-speed rail line, capable of carrying "bullet trains" that travelled at around 210 km/h (130 mph). Some trains today can travel even faster.



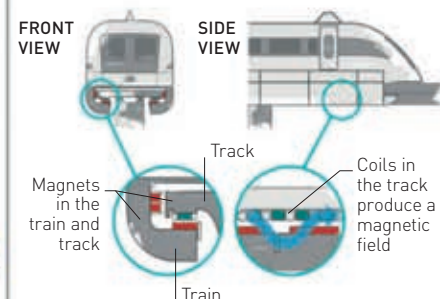
MAGNETIC TRAINS

Maglev trains use magnetic levitation to move trains along without having to touch the ground or any form of rail. Magnets allow them to rise and then travel extremely fast above special rails.



MAGLEV TRAIN

HOW MAGLEV WORKS



1800

1804 Trevithick's Penydarren locomotive is tried in South Wales.

TRAIN HISTORY

Trains have changed significantly over their 300-year history. The first trains were steady and steam-powered. Modern trains are usually electric and can reach very high speeds.



London Underground map

1812 First successful commercial use of steam, in Yorkshire, England.

1829 *Stephenson's Rocket* becomes the fastest train when it reaches 46 km/h (29 mph).

1863 Underground railway opens in London, UK.

1869 The transcontinental railway opens, making the USA a dominant economic power. Railways create big business in America, driving the Industrial Revolution.

1913 Sweden uses the first diesel-powered main line trains.

1890 First use of electric locomotives on London's underground.

1914-18 Large armies are moved by train during World War I.

1960s USA phases out steam engines.



East German Battery shunter 1966

1964 Japan introduces a super-fast "bullet" train.



Japanese bullet train

2000s

2014 Japan's Maglev trains run faster than 480 km/h (300 mph).